Data Quality Findings

1. Dataset Overview

The assessment was performed on a retail lending dataset containing 10,000 loan records and 55 variables, comprising 42 numeric and 13 categorical variables. The dataset provides sufficient observations and variable coverage to support downstream IFRS 9 model development activities, including PD, LGD, and EAD modelling.

2. Missing Value Assessment

The majority of variables exhibit minimal missing values. Variables with the missing percentages larger than 5% are shown as below:

No.	Variable	Missing %	Assessment
1	verification_income_joint	85.45%	Review
2	annual_income_joint	85.05%	Review
3	debt_to_income_joint	85.05%	Review
4	months_since_90d_late	77.15%	Review
5	months_since_last_delinq	56.58%	Review
6	months_since_last_credit_inquiry	12.71%	Review
7	emp_title	8.33%	Acceptable
8	emp_length	8.17%	Acceptable

Business Interpretation

The high missingness observed in the joint application variables is attributable to the predominance of individual loan applications. When restricted to joint applications, the missing rates are negligible (verification_income_joint: 2.68%; annual_income_joint: 0% and debt_to_income_joint: 0%), indicating that these fields are generally complete when applicable.

The high missingness in months_since_90d_late and months_since_last_delinq makes sense as 85.76% of the borrowers have no delinquency or delinquency in recent 2 years.

The remaining variables exhibit relatively low missingness and do not represent material concerns.

Recommended Cleaning Actions

- Replace missing values or NA with "Unknown" category for optional variables such as emp_title and emp_length during feature engineering.

- Investigate months_since_last_credit_inquiry to determine whether missing values represent unavailable bureau information or valid business scenarios.

3. Duplicate Record Assessment

No duplicate records were identified.

Business Interpretation

The absence of duplicate observations indicates that each loan record appears to represent a unique exposure.

Recommended Cleaning Action

No action required.

4. Business Rule Validation

All critical validation rules passed successfully.

The following two business rules were flagged for review:

No.	Validation Rule	Failed %	Assessment
1	Credit Utilized ≤ Credit Limit	1.54%	Review
2	CC Carrying Balance Accounts ≤ Open CC Accounts	0.10%	Review

Business Interpretation

The 154 observations where total_credit_utilized exceeds total_credit_limit are likely attributable to legitimate business scenarios, such as temporary over-limit balances caused by accrued interest, fees, or reporting timing differences. These observations are considered plausible and do not necessarily indicate invalid data.

The 10 observations where the number of credit card accounts carrying a balance exceeds the number of open credit card accounts warrant further investigation. While the occurrence rate is very low (0.10%), this relationship is logically inconsistent and may reflect data recording or reporting issues.

Recommended Cleaning Actions

- Retain records where credit utilization exceeds the reported credit limit.
- Review the 10 inconsistent credit card account records to confirm whether they represent genuine data quality issues. Correct or exclude these records if inconsistencies cannot be resolved.

5. Distribution & Extreme Value Assessment

The distribution analysis indicates that many financial and behavioural variables exhibit substantial positive skewness, heavy tails, or a high proportion of zero values. These characteristics are typical of retail consumer lending portfolios rather than indicators of poor data quality. The following observations were identified:

Highly Skewed Financial Variables

Several continuous financial variables exhibit strong positive skewness, including:

- annual_income
- total_credit_limit
- total_credit_utilized
- paid_total
- paid_principal
- total_debit_limit

The histogram and boxplot indicate long right tails driven by a relatively small number of borrowers with substantially higher balances or incomes than the remainder of the portfolio.

Business Interpretation

These distributions are expected within consumer lending portfolios and reflect natural differences in borrower financial capacity rather than data anomalies.

Recommended Action

Retain these variables in their original form during the data quality stage. Evaluate whether transformations (e.g., logarithmic transformation or winsorization) improve model performance during feature engineering.

Zero-Inflated Variables

Several variables contain a very high proportion of zero values, including:

- total_collection_amount_ever
- num_collections_last_12m
- tax_liens
- paid_late_fees
- current_accounts_delinq
- num_accounts_30d_past_due
- num_accounts_120d_past_due

Business Interpretation

These variables represent relatively rare credit events. The predominance of zero values reflects borrowers with no adverse credit history and is expected in a performing retail loan portfolio.

Recommended Action

Retain these variables. Consider deriving binary indicators (e.g., "Has Collection History") during feature engineering if appropriate.

6. Variable Variability Assessment

The majority of variables demonstrate sufficient variability to support statistical modelling.

Three variables exhibit near-zero variance:

No.	Variable	Assessment
1	current_accounts_delinq	Near-zero variance
2	num_accounts_120d_past_due	Near-zero variance
3	num_accounts_30d_past_due	Near-zero variance

Additionally, three variables are highly concentrated:

No.	Variable	Assessment
1	num_collections_last_12m	Highly concentrated
2	tax_liens	Highly concentrated

3	paid_late_fees	Highly concentrated

Business Interpretation

These variables represent rare adverse credit events.

The limited variability is expected because the vast majority of borrowers have:

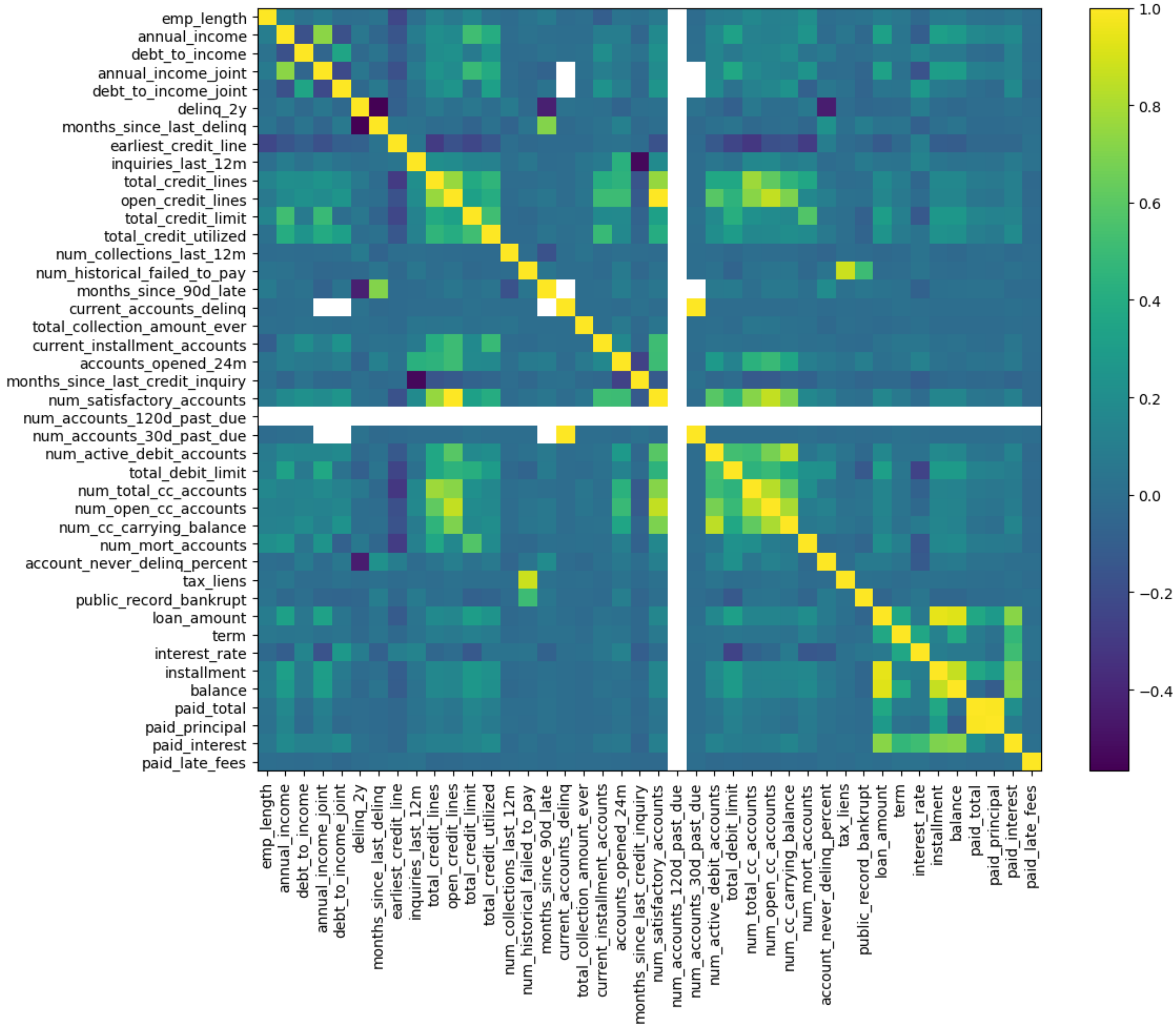
- no current delinquencies,
- no tax liens,
- no recent collections,
- no late payment fees.

Although these variables exhibit limited statistical variation, they may remain valuable predictors of default because the occurrence of these events is strongly associated with increased credit risk.

Recommended Action

Retain these variables during feature engineering.

7. Correlation Assessment



The correlation analysis identified several groups of variables exhibiting moderate to strong positive relationships. Notable correlations include:

Credit Exposure Variables

- total_credit_limit
- total_credit_utilized
- open_credit_lines
- total_credit_lines
- num_satisfactory_accounts

These variables measure related aspects of borrower credit exposure and naturally exhibit positive correlation.

Credit Card Account Variables

Moderate correlation exists between:

- num_total_cc_accounts
- num_open_cc_accounts
- num_cc_carrying_balance
- total_debit_limit

These relationships are expected because the variables describe different aspects of the same credit card portfolio.

Loan Performance Variables

Strong positive correlations are observed among:

- loan_amount
- installment
- balance
- paid_total
- paid_principal

These relationships are expected because they are mathematically and economically linked throughout the life cycle of a loan.

Low Correlation Variables

Several rare-event variables demonstrate weak or negligible correlations with the remainder of the portfolio, including:

- num_accounts_120d_past_due
- current_accounts_delinq
- tax_liens

This behaviour is expected because these variables exhibit near-zero variance and represent infrequent credit events.

Business Interpretation

The observed correlations primarily reflect expected financial relationships rather than redundant or erroneous data.

While several variables exhibit relatively high pairwise correlations, this does not represent a data quality issue. Instead, these relationships should be considered during feature selection to minimise multicollinearity in statistical models.

Recommended Action

- Retain correlated variables during the data quality stage.
- Assess multicollinearity using during feature engineering.
- Select the most predictive variables where highly correlated measures provide similar information.

Overall Data Quality Conclusion

The dataset demonstrates a high level of completeness, consistency, and integrity suitable for IFRS 9 Expected Credit Loss model development. No duplicate records or critical business rule violations were identified. High missingness is largely attributable to conditionally applicable variables, while skewed distributions, zero-inflated variables, and near-zero variance variables are consistent with expected characteristics of retail lending portfolios. The dataset is considered fit for purpose, with only minor data preparation activities required prior to feature engineering and model development.